

HALLI, Finland

WMO: 029450

Lat: 61.855N

Lon: 24.805E

Elev: 475

StdP: 14.45

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-15.0	-8.9	-19.9	1.9	-15.1	-12.9	2.8	-8.8	20.9	33.1	18.0	31.4	2.6	320	0.619

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	80.3	63.9	76.9	62.4	73.6	60.9	66.7	75.5	64.8	73.1	62.9	70.5	7.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	90.9	69.5	61.7	83.7	67.3	59.9	78.3	65.8	31.5	75.0	30.0	72.9	28.7	70.4	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.6	15.2	13.4	DB	-21.0	84.0	7.5	3.6	-26.4	86.5	-30.8	88.6	-35.0	90.6	-40.4	93.2	(4)
(5)			WB	-21.0	69.0	7.4	2.7	-26.3	70.9	-30.6	72.5	-34.8	74.0	-40.2	76.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	18.9	18.7	26.1	37.6	48.5	57.0	62.4	59.4	50.1	39.6	31.3	24.1		
	DBStd	17.48	12.68	12.77	8.80	6.58	7.83	6.44	5.84	5.68	6.24	7.26	8.07	11.86		
	HDD50	4862	964	878	740	377	123	11	0	3	75	326	561	804		
	HDD65	9345	1429	1298	1205	823	513	252	122	190	446	787	1011	1269		
	CDD50	1061	0	0	0	3	77	220	385	293	79	4	0	0		
	CDD65	71	0	0	0	0	2	12	42	15	0	0	0	0		
	CDH74	646	0	0	0	0	36	127	370	113	0	0	0	0		
	CDH80	98	0	0	0	0	3	14	71	10	0	0	0	0		
	WSAvg	6.1	5.9	5.9	6.5	6.1	6.5	6.4	5.6	5.5	5.9	6.2	6.3	6.4		
	PrecAvg	26.1	2.1	1.5	1.4	1.3	1.5	2.7	3.5	3.0	2.1	2.7	2.2	2.0		
	PrecMax	35.4	3.6	3.0	2.7	2.2	3.7	4.5	5.7	6.2	5.3	5.2	4.9	3.7		
	PrecMin	22.2	0.7	0.1	0.3	0.3	0.2	1.1	0.7	0.9	0.5	0.5	0.2	0.6		
	PrecStd	2.8	0.8	0.8	0.6	0.6	0.8	1.0	1.3	1.5	1.2	1.3	1.2	0.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.9	40.2	49.8	65.8	78.4	82.1	85.5	81.0	70.0	57.1	48.7	44.3		
		MCWB	38.0	36.6	40.4	51.2	59.2	63.7	66.7	64.3	60.0	52.5	47.7	43.1		
	2%	DB	36.5	37.1	43.9	58.9	72.8	77.7	81.8	77.0	66.0	53.4	46.1	41.2		
		MCWB	35.7	35.6	37.4	46.9	56.7	60.9	65.7	63.4	58.2	51.0	45.0	40.6		
	5%	DB	34.7	34.9	39.9	53.9	68.0	73.7	78.2	73.9	62.5	51.7	44.0	38.9		
		MCWB	34.2	34.0	35.3	44.1	54.1	59.3	63.8	61.7	56.3	49.8	43.3	38.1		
	10%	DB	33.2	33.2	37.0	49.9	63.3	70.0	74.6	70.6	59.6	49.8	41.4	36.5		
		MCWB	32.8	32.4	34.0	42.1	51.6	57.6	62.5	60.5	55.1	48.1	40.7	35.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.8	37.6	41.8	52.5	61.8	66.8	70.0	68.1	61.3	53.9	47.9	43.2		
		MCDB	38.0	38.8	47.7	60.2	73.2	76.8	79.0	76.5	66.6	55.8	48.6	43.8		
	2%	WB	35.8	35.7	38.2	48.5	58.6	63.7	67.7	65.3	59.3	51.8	44.9	40.6		
		MCDB	36.1	36.8	42.0	56.2	68.2	72.7	77.1	72.7	63.4	52.9	45.5	41.0		
	5%	WB	34.1	34.1	36.2	45.4	55.9	61.4	65.9	63.5	57.8	50.2	43.0	38.0		
		MCDB	34.5	34.9	39.0	52.9	65.6	70.0	74.4	70.5	61.4	51.5	43.6	38.5		
	10%	WB	32.6	32.4	34.4	42.6	52.9	59.2	64.1	61.9	55.8	48.2	40.9	35.8		
		MCDB	32.9	33.0	36.4	48.7	61.7	67.8	72.3	68.4	58.9	49.3	41.3	36.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.1	11.4	14.3	16.4	18.6	17.0	16.7	16.6	13.7	8.9	6.9	8.4		
		MCDBR	7.3	8.6	16.8	24.5	26.7	23.6	22.3	21.8	17.1	10.0	7.6	7.1		
		MCWBR	7.2	8.0	11.7	14.5	12.8	10.4	9.0	9.7	10.2	7.9	7.5	7.0		
	5% WB	MCDBR	7.0	8.1	13.8	21.8	22.9	19.7	18.9	18.0	14.5	9.0	7.5	6.9		
		MCWBR	7.0	7.8	10.2	13.7	12.0	10.1	8.9	9.3	9.5	8.0	7.4	7.0		
(40) Clear-Sky Solar Irradiance	taub		0.242	0.289	0.317	0.351	0.349	0.346	0.369	0.355	0.332	0.316	0.264	0.356		
	taud		2.063	2.153	2.231	2.374	2.455	2.487	2.438	2.470	2.527	2.417	2.162	2.614		
	Ebn at Noon		140	211	250	266	277	279	268	263	249	206	132	91		
	Edh at Noon		12	22	29	31	31	30	31	28	22	17	10	7		
(44) All-Sky Solar Radiation	RadAvg		68	244	711	1162	1562	1654	1591	1233	715	297	78	35		
	RadStd		12	57	87	128	175	130	162	141	93	55	15	4		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	-254	N/A	N/A	N/A

Nomenclature: See separate page